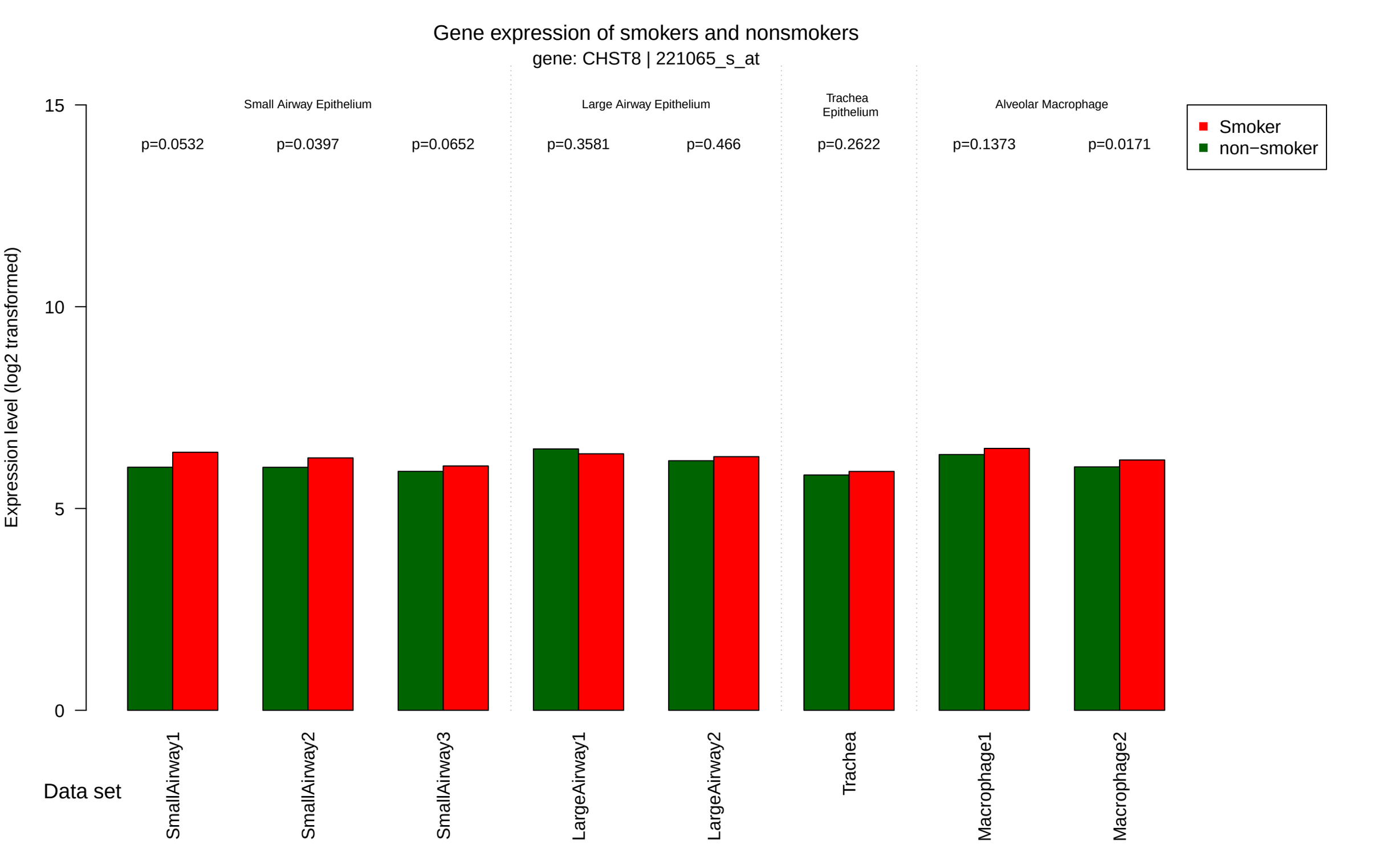
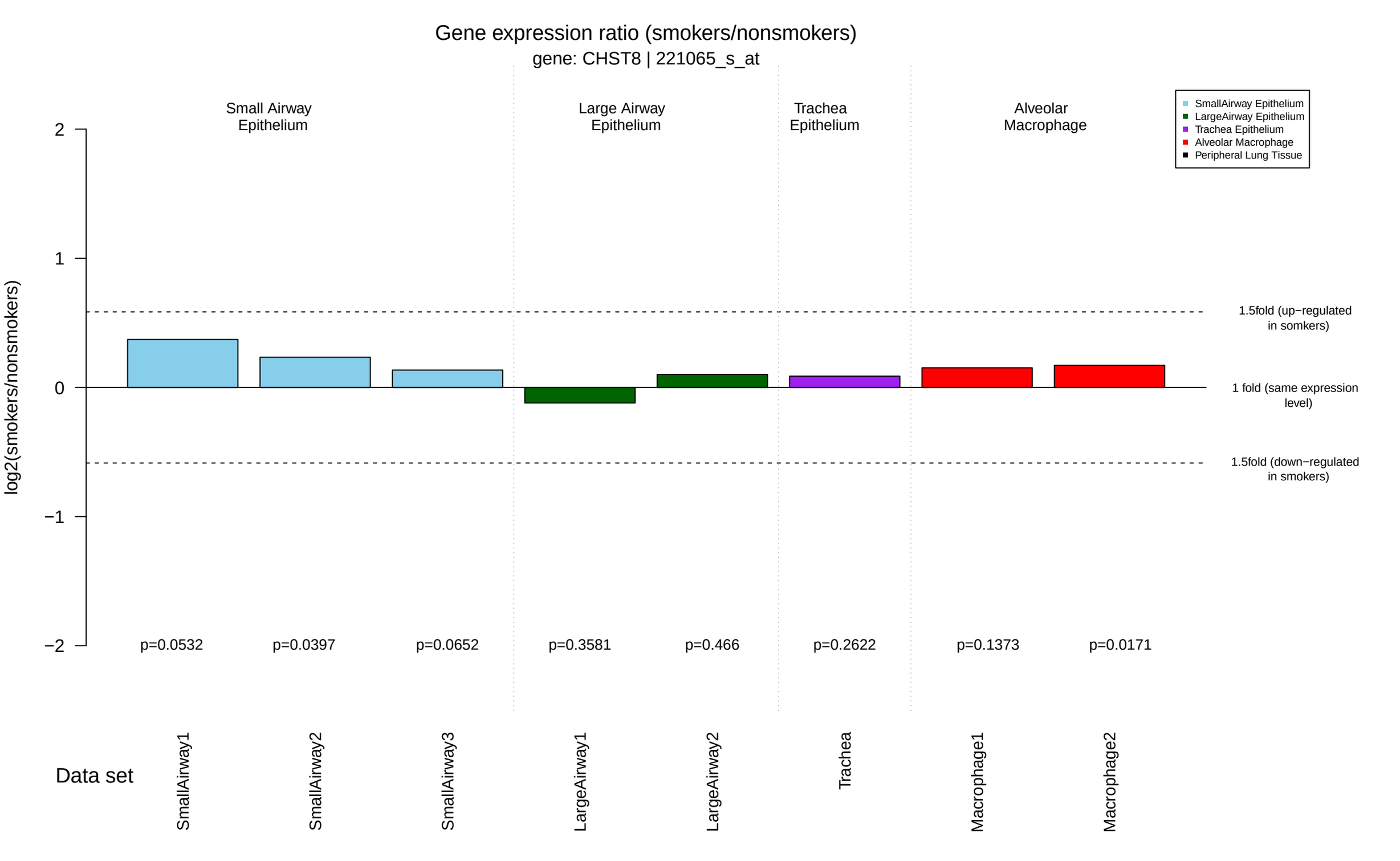
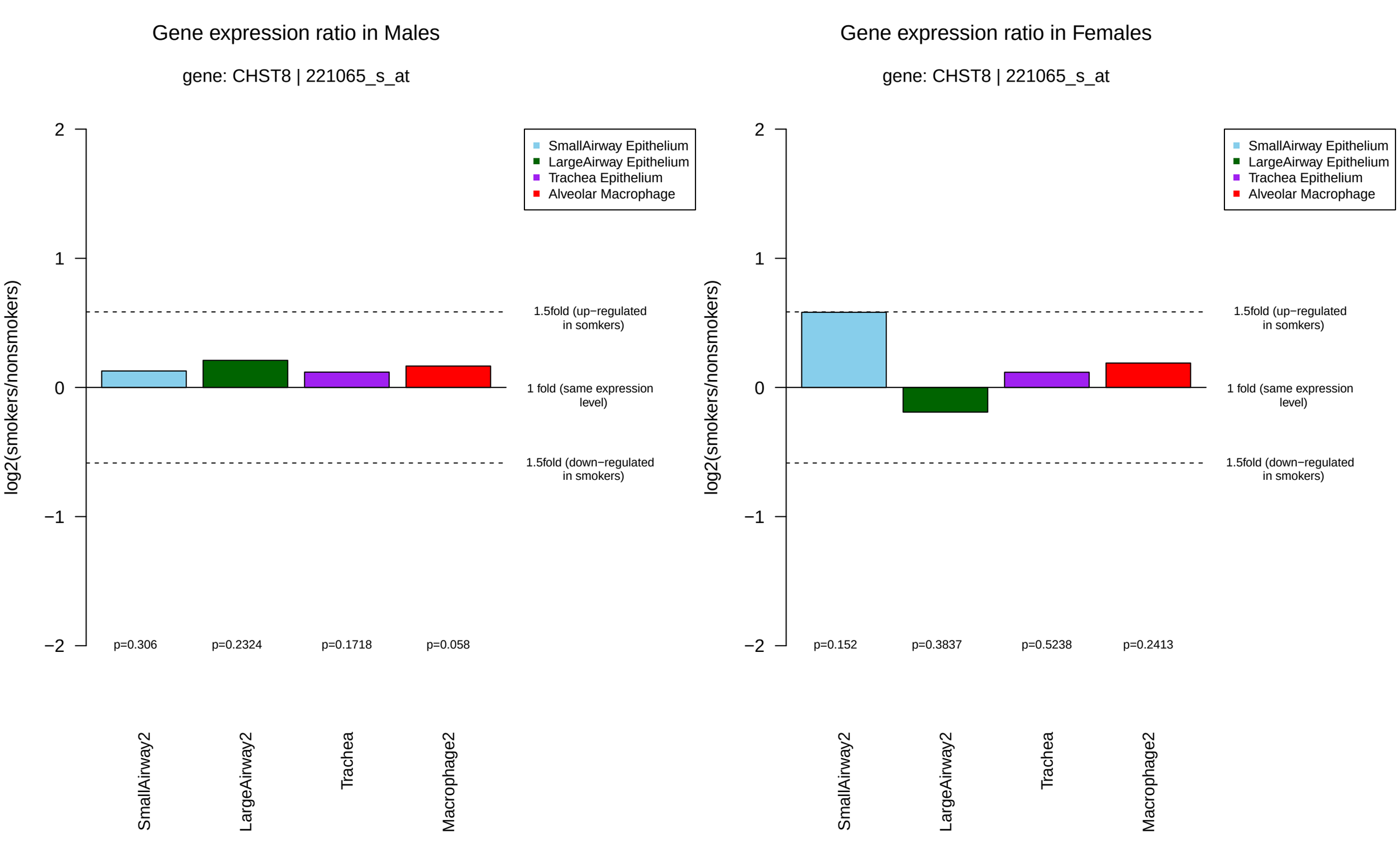
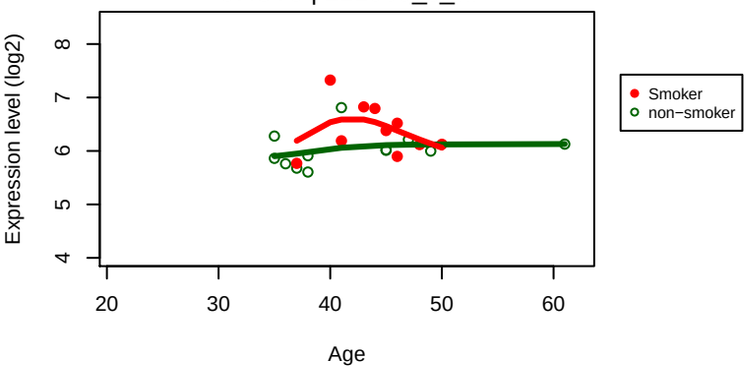
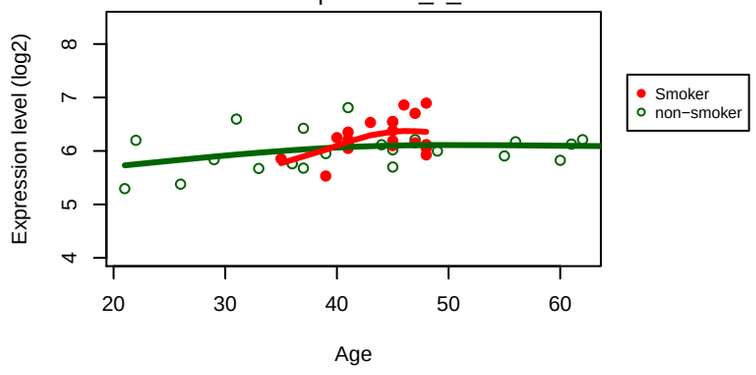
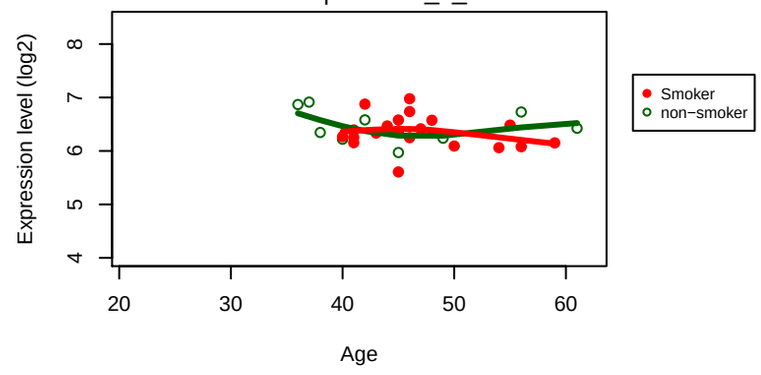

Gene expression of smokers and nonsmokers
gene: CHST8 | 221065_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0532
p=0.0397
p=0.0652
p=0.3581
p=0.466
p=0.2622
p=0.1373
p=0.0171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

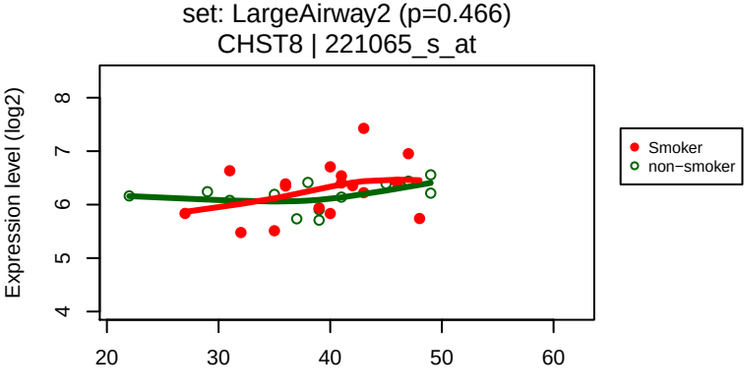

Gene expression ratio (smokers/nonsmokers)
gene: CHST8 | 221065_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0532
p=0.0397
p=0.0652
p=0.3581
p=0.466
p=0.2622
p=0.1373
p=0.0171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

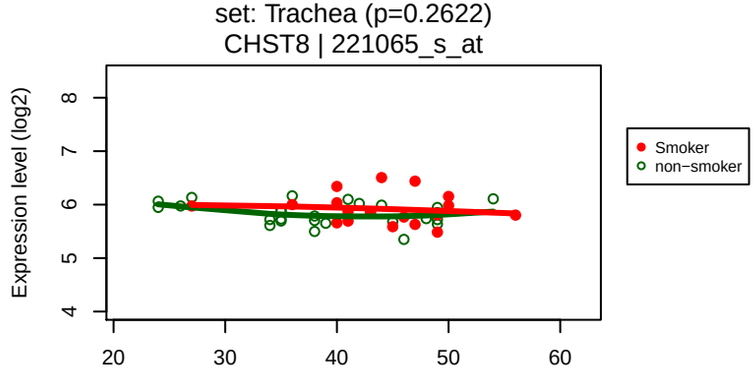

Gene expression ratio in Males
gene: CHST8 | 221065_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.306
p=0.2324
p=0.1718
p=0.058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST8 | 221065_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.152
p=0.3837
p=0.5238
p=0.2413
SmallAirway2
LargeAirway2
Trachea
Macrophage2

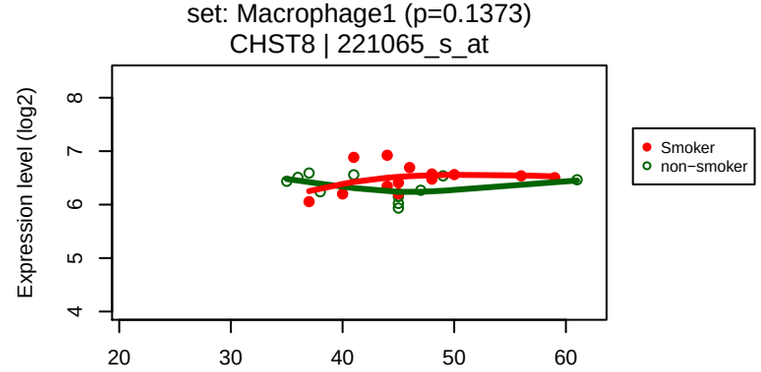

set: SmallAirway1 (p=0.0532)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker

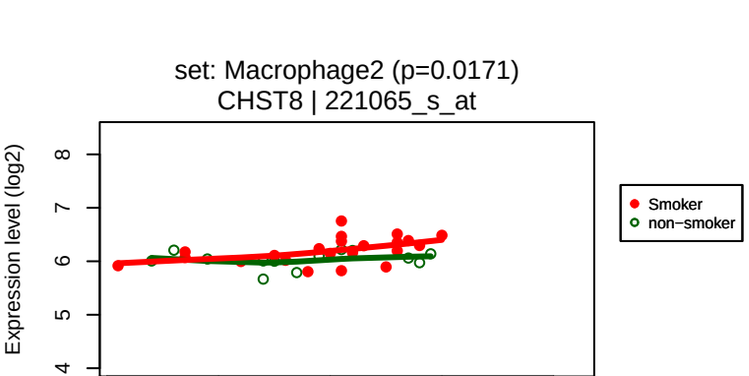

set: SmallAirway2 (p=0.0397)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker

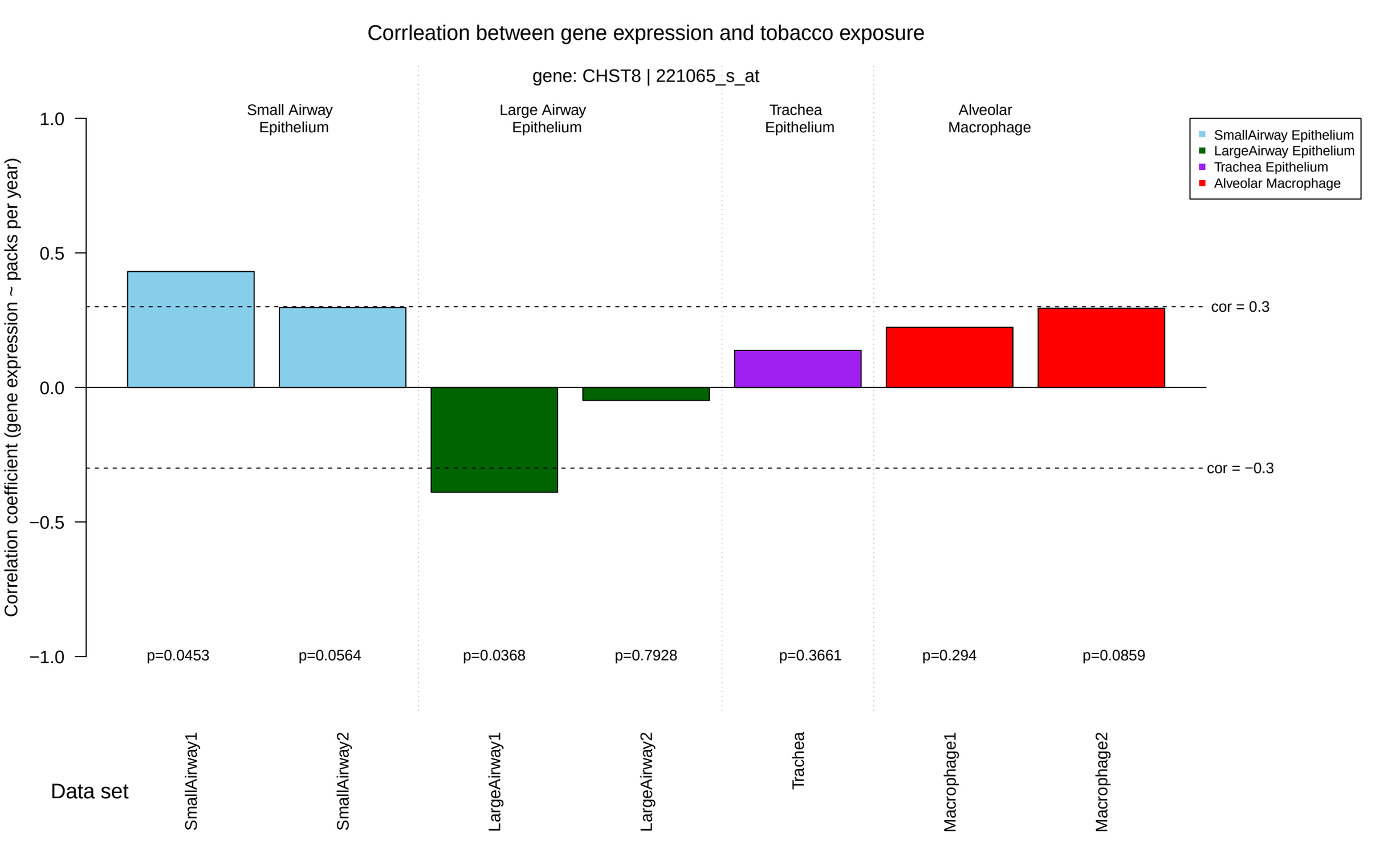

set: LargeAirway1 (p=0.3581)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.466)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2622)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1373)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0171)
CHST8 | 221065_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST8 | 221065_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0453
p=0.0564
p=0.0368
p=0.7928
p=0.3661
p=0.294
p=0.0859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2